

Fibonacci Numbers

$$F_0 = 0$$

$$F_{n+2} = F_n + F_{n+1}$$

$$F_1 = 1$$

$$0, 1, 1, 2, 3, 5, 8, \dots$$

Lemma: For all n , $F_n < 2^n$

Proof by induction.

Base Case: $n=0$

$$F_0 = 0 \text{ and } 2^0 = 1 \text{ so } F_0 < 2^0$$

$$n=1 \quad F_1 = 1 \text{ and } 2^1 = 2 \text{ so } F_1 < 2^1$$

Inductive Step: Assume for n

then $F_n < 2^n$ Also for $n-1$

$$\text{so } F_{n-1} < 2^{n-1}$$

By hypothesis $F_{n+1} = F_{n-1} + F_n$

$$F_{n-1} + F_n < 2^{n-1} + 2^n = 3 \cdot 2^{n-1} < 4 \cdot 2^{n-1} = 2^{n+1}$$

$$F_{n+1} = F_n + F_{n-1} < 2^{n-1} + 2^n < 2^n + 2^n = 2^{n+1} \text{ so } F_{n+1} < 2^{n+1} \text{ QED}$$

$0 < 1$ and $1 < 2 \Rightarrow 0+1 < 4$

also $3 < 4$

add $<$

$$\begin{array}{r} 0 < 1 \\ 1 < 2 \\ \hline 1 < 3 \end{array}$$

Lemma: $\forall n \geq 0 \quad \sum_{i=1}^n F_i^2 = F_n F_{n+1}$

Proof by induction:

Base Case: $n=1$

$$\sum_{i=1}^1 F_i^2 = F_1^2 = 1^2 = 1$$

$$F_1 F_{1+1} = F_1 F_2 = 1 \cdot 1 = 1$$

$$\left(\begin{array}{c} F_{n+1}^2 \\ + \\ F_n F_{n+1} \end{array} \right) = F_{n+1}^2 + F_n F_{n+1}$$

Inductive Step:

Assume true for n

$$\text{So } \sum_{i=1}^n F_i^2 = F_n F_{n+1}$$

We have to show $\sum_{i=1}^{n+1} F_i^2 = F_{n+1} F_{n+2}$

To get started. Rest is algebra

$$F_{n+2} = F_n + F_{n+1} \quad (\text{Given})$$

$$\sum_{i=1}^{n+1} a_i = a_{n+1} + \sum_{i=1}^n a_i \quad (\text{Fact about Sums})$$

